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SUPERSEDING
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MILITARY SPECIFICATION

TIMER, INTERVAL, 115 volt, 60 Hz, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electrically powered timer suitable for accurately indicating intervals of time.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-packaging, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila. PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6645

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 - Dimensions of Plugs and Receptacles--General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Ave., Boston, MA 02210.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 863 - Electric Time-Indicating and Recording Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, NY 11749; 207 East Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

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3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 863 and NFPA No. 70. Proof shall consist of one of the following:

a. UL label, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified.

3.1.2 Compliance with above paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Material. Material shall be of the highest quality throughout, suitable in all respects for the purpose intended, and as specified herein.

3.3 Construction.

3.3.1 Housing. The sides, base and top shall be cold rolled steel. The front panel shall be anodized aluminum or steel and have openings which provide access to the digital dials (see 3.3.2). Housing shall be mounted on supports. Supports shall be higher in front than they are in back so that the front panel is inclined (see table I).

3.3.2 Dials. The timer shall have 3 digital dials (numbered wheels) capable of being individually pre-set to indicate any time interval from 99.9 to 00.0 minutes. A plastic cover may be secured in the area of the dials provided it does not interfere with dial movement, does not obstruct the view of graduations or move while

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timer is being used. The dials shall be so placed that they combine to form readable numbers such as 23.5. The left hand and middle dials shall be graduated and marked in steps of 1 from 9 to 0 The right hand dial shall be graduated and marked so that it indicates in steps of .1 from .9 to .0. Decimal point shall be marked on the dial or on the plastic cover.

3.3.3 Clutch mechanism. The timer shall have a undirectional clutch mechanism to prevent time loss or inaccuracy of the digital dials. A lever shall be mounted on the front panel or on the top to disengage the clutch mechanism to permit the individual setting of each dial.

3.3.4 Alarm. The timer shall be equipped with a buzzer alarm that sounds for 3 seconds when there is 01.0 minute remaining on the timer and again at 00.0 minutes when the pre-set time cycle has been completed, but does not sound at any other time. The alarm shall be audible at a distance of 50 feet (maximum setting - see 3.3.4.1).

3.3.4.1 Volume control. A volume intensity control regulator for the alarm may be supplied. If supplied it shall be located and identified on the rear panel of the timer.

3.3.5 Power switch. The timer shall operate on a nominal 115 volts, 60 Hz alternating current. The timer shall have an "off-on" electrical switch mounted on the front panel. Following the completion of the pre-set time cycle at 00.0 minutes, the timer mechanism shall initiate a new time cycle and indicate intervals of time elapsed until power switch is turned off.

3.3.6 Cord and plug assembly. The timer shall be fitted with a well-insulated, flexible, heavy-duty, 3-wire power cord. The cord shall be at least 6 feet long when measured from the outside of the back panel and shall be fitted with a ground-type plug, conforming to ANSI C73.11.

3.4 Dimensions, weights and angles. Dimensions, weights and angles shall be as shown in table I.

TABLE I. Dimensions, weights and angles.

	Dimensions, weights and angles	Tolerance
Width	4-5/8 in.	± 3/8 in.
Height	4-3/8 in.	± 3/8 in.
Depth	4-1/2 in.	± 1/4 in.
Weight	3-1/4 lbs.	maximum
Thickness of steel for housing	.036 in.	minimum
Front panel inclination	8-1/4°	± 1-3/4°

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3.5 Accuracy. Timer shall read the same as the standard ± 6 seconds, when tested as specified in 4.4.1.

3.6 Finish. The external housing shall be vinyl clad or shall have a durable, baked enamel finish. Where consistent with the use and proper functioning of parts, all other parts of the timer shall be finished or treated so as to resist corrosion.

3.7 Operating instructions. Operating instructions shall be legibly and permanently marked on the rear of the housing.

3.8 Identification marking. Each timer shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.9 Workmanship. Workmanship shall be first class throughout. The timer shall be free of defects listed in table IV and any other defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

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* TABLE II. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual and audio examination		
Major A defects	S-2	1.0
Major B defects	II	1.0
Major C defects	II	2.5
Minor defects	II	4.0
For dimensional, weight, and angular examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. Acceptance number shall be zero.

TABLE III. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Timer	Accuracy	3.5	4.4.1	S-1

4.3 Examination. The timer shall be examined to determine compliance with all requirements in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in table IV.

* TABLE IV. Classification of defects.

Category	Defect
Major A	
101	Alarm not audible at 50 feet.
102	Alarm does not sound at specified times.
103	Alarm sounds at unspecified times.
Major B	
111	Component missing.
112	Graduation markings, numbers and decimal p ints not complete, correct, legible or permanent.

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TABLE IV. Classification of defects.(cont'd)

Category	Defect
Major C	
121	Component not secured.
122	Timer not free of dents and deformation.
123	Finish not smooth and uniform.
124	Timer not stable when base is placed on a flat surface.
125	Operating instructions and volume control markings not complete, correct, legible or permanent.
Minor	
201	Timer not free of dirt, grease or foreign matter.
202	Identification marking not complete, correct, legible or permanent.

4.3.1 Dimensional, weight and angular examination. Timer shall be examined for defects in dimensions and weight. In addition, angle of front panel shall be measured. Any dimension, weight or angle not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The timer shall be run on 115 volt, 60 Hz alternating current using an electrical timer as a standard. Comparison with standard shall be made at the end of 100 minutes.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

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TABLE V. Examination for preparation for delivery.

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit preparation. Each unit shall be cleaned and dried in accordance with the applicable method as specified in MIL-P-116. A preservative shall not be utilized on the item, and each item shall be wrapped in neutral tissue or other suitable non-corrosive material.

5.1.1.2 Unit package. Each timer shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

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5.2.1 Level A.

5.2.1.1 Exterior container. Eighteen unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements as specified in the applicable table of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to the applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Eighteen unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Special requirements as specified in the applicable table of PPP-B-636 shall apply for fiberboard boxes. Closure of fiberboard boxes shall conform to method I or II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to the point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6645-00-089-6156	TIMER, INTERVAL, 115 volt, 60 Hz, AC

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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